Regression Models in Practice: Model Capacity, Gradient-Descent Logistic Regression, and L1/L2 Regularisation

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Abstract

We study three practical questions in regression, each on its own dataset. First, capacity*: on 100 samples from a noisy linear model, polynomials of degree 2 and 3 lower the training error but not the test error (test MSE 0.654, 0.636 and 0.642 for degrees 1–3), so the extra terms only fit noise.*

Second, optimisation*: logistic regression trained with hand-written batch gradient descent on a 200-sample, imbalanced binary dataset. With learning rates 0.01 and 0.05 and the 20-iteration budget, the model never beats the 74% majority-class rate. Run to convergence it reaches 88%, so the budget, not the model, is the bottleneck.*

Third, regularisation*: ordinary least squares, LASSO and ridge on a 10-feature dataset. Mapping the textbook penalty $\lambda/2 = 1$ onto scikit-learn's scaling matters a great deal. Correctly mapped, LASSO is mild (test MSE 4.64, all coefficients kept). Our first implementation used a penalty 160× stronger, which zeroed 6 of 10 coefficients and raised the test MSE to 21.8.*

1. Introduction

Fitting a regression model involves three choices that are easy to get subtly wrong: how flexible the model should be, how it is optimised, and how strongly it is penalised. We study each with a small, controlled experiment and report what the numbers do and do not support. All three use an 80/20 train/test split with seed 42.

2. Related Work

The bias–variance trade-off [2] explains why extra flexibility can hurt on unseen data. Logistic regression [1] is the canonical probabilistic linear classifier, and gradient descent its simplest optimiser. Ridge regression [3] shrinks coefficients with an L2 penalty, while the LASSO [4] uses an L1 penalty that sets some coefficients exactly to zero, performing variable selection.

3. Method

3.1. Polynomial regression

We draw $x \sim \mathcal{U}(-3, 3)$ and set $y = 2x - 1 + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, 1)$, for $N = 100$ samples. Polynomials of degree 1, 2 and 3 are fitted by least squares on the training 80, and the MSE is computed on the held-out 20.

3.2. Logistic regression by gradient descent

The dataset has 200 samples, two features (one in $[0.1, 10]$, one in the tens of thousands) and a binary label with 26% positives. Features are standardised and a bias column is added; the row index column is excluded. Starting from $w = 0$, we apply 20 full-batch updates

$$w \leftarrow w - \eta \, \tfrac{1}{N} X^\top (\sigma(Xw) - y), \tag{1}$$

which is the gradient of the mean cross-entropy, with $\eta \in \{0.01, 0.05\}$. At each iteration we track the MSE between $\sigma(Xw)$ and y, the cross-entropy and the accuracy.

3.3. OLS vs. LASSO vs. ridge

The dataset has 100 samples, 10 standardised features and a continuous target. With the regularised error $E = \sum_t (r^t - g(x^t))^2 + \lambda\,\Omega(w)$, we use $\lambda/2 = 1$ for the L1 penalty and $\lambda/2 = 0.1$ for the L2 penalty.

Matching the penalty to the library. scikit-learn scales the two penalties differently. Ridge minimises $\mathrm{RSS} + \alpha\|w\|_2^2$, so $\alpha = \lambda = 0.2$. Lasso minimises $\frac{1}{2n}\mathrm{RSS} + \alpha\|w\|_1$, so $\alpha = \lambda/(2n) = 1/80$ for $n = 80$ training samples. Our first implementation passed $\alpha = 2$ directly, a 160× stronger penalty; we keep it as a fourth model for comparison. If E were written with the *mean* squared error instead, the matching value would be $\alpha = 1$; Fig. 4 covers the whole range. All models are fitted on the same standardised features so their coefficients are comparable.

4. Results and Discussion

4.1. Capacity: degree 1 is enough

Training MSE falls with every added degree, as it must, but test MSE changes by less than 0.02 (Table 1). With only 20 test points that difference is well within noise. The fitted

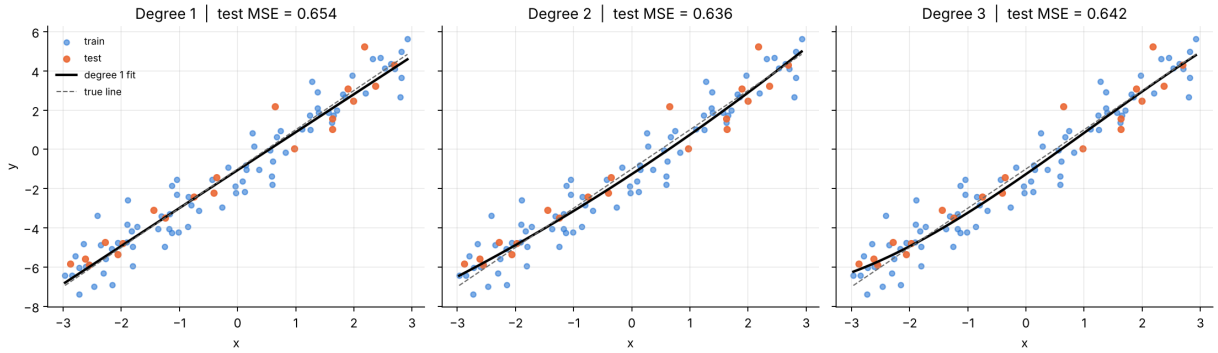


Figure 1. Polynomial fits of degree 1–3 to 100 noisy samples of $y = 2x - 1$ (dashed). The higher-degree curves are almost indistinguishable from the line.

Table 1. Polynomial regression. Coefficients are listed from the linear term upward.

Degree	Train MSE	Test MSE	Intercept; coefficients
1	0.848	0.654	$-1.058;\ 1.933$
2	0.815	**0.636**	$-1.264;\ 1.943,\ 0.067$
3	**0.808**	0.642	$-1.263;\ 2.056,\ 0.067,\ -0.021$

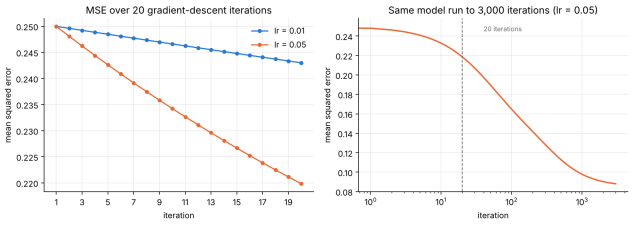


Figure 2. Left: MSE over 20 iterations for both learning rates. Right: the same model run for 3,000 iterations ($\eta = 0.05$); the dashed line marks iteration 20.

Table 2. Logistic regression after gradient descent. Weights are $[b, w_1, w_2]$ on standardised features; CE is cross-entropy.

Setting	Final weights	MSE	CE	Acc.
η 0.01, 20 it.	$[-0.047,\ 0.021,\ -0.017]$	0.243	0.679	74.5%
η 0.05, 20 it.	$[-0.214,\ 0.102,\ -0.086]$	0.219	0.632	74.5%
η 0.05, 3000 it.	$[-1.77,\ 3.51,\ -3.41]$	0.088	–	**88.0%**

quadratic and cubic coefficients are near zero, and the linear fit recovers the true slope (1.93 vs. 2) and intercept (-1.06 vs. -1). All three are close to the irreducible noise variance of 1. The data are linear, and the extra terms add variance without reducing bias.

4.2. Gradient descent: the budget is the bottleneck

Both learning rates reduce the MSE steadily and almost linearly (Fig. 2), and $\eta = 0.05$ reduces it five times faster per step. After 20 steps the weights are still tiny: the model has learned the class imbalance (negative bias) and the sign of each feature, but its 74.5% accuracy matches the 74% majority-class rate. Continuing to 3,000 iterations raises accuracy to 88%, with weights an order of magnitude larger (Table 2). On standardised features this convex problem could safely use a much larger learning rate.

The two features push in opposite directions ($w_1 > 0$, $w_2 < 0$), consistent with their correlations with the label ($+0.24$ and -0.20). An earlier version also fed the row index in as a feature. Its weight was near zero, but it carries no information and is excluded here.

Table 3. Regression with and without regularisation (20 test samples).

Model	sklearn α	Test MSE	Non-zero coef.
OLS	–	4.722	10 / 10
LASSO, $\lambda/2 = 1$	1/80	**4.640**	10 / 10
Ridge, $\lambda/2 = 0.1$	0.2	4.690	10 / 10
LASSO, over-strong	2	21.787	4 / 10

4.3. Regularisation: strength is everything

At the specified strengths both penalties are mild. LASSO and ridge change the coefficients by a few percent and lower the test MSE slightly, from 4.72 to 4.64 and 4.69 (Table 3, Fig. 3). With 80 training samples and 10 features, OLS is not badly overfit, so there is little for regularisation to fix.

The over-strong LASSO shows the other side of the trade-off. It keeps only the four strongest features (x_0, x_1, x_2, x_9), shrinks even those by 30–60%, and its test MSE rises 4.6-fold: classic underfitting. The path in Fig. 4 makes the progression clear. Test error is flat up to $\alpha \approx 0.1$ (with a shallow minimum of 4.24 at $\alpha = 0.1$, where 2 coefficients are dropped), then climbs steeply as useful features are removed.

5. Conclusion & Future Work

On linear data a linear model generalises as well as a cubic one; 20 gradient steps leave logistic regression far from its optimum; and whether a regulariser helps or harms depends entirely on its strength, which in turn depends on matching textbook and library conventions.

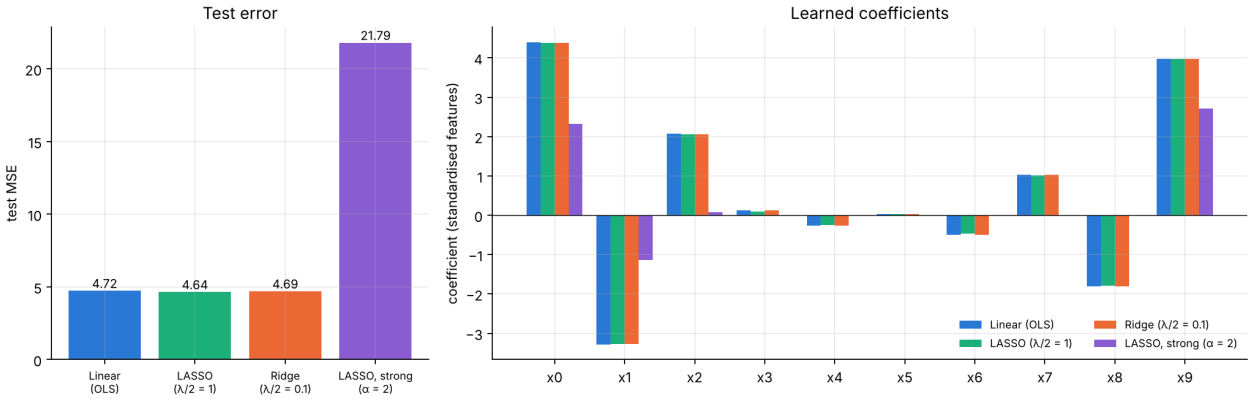


Figure 3. Test MSE (left) and learned coefficients on standardised features (right) for OLS, correctly mapped LASSO and ridge, and the over-strong LASSO with $\alpha = 2$.

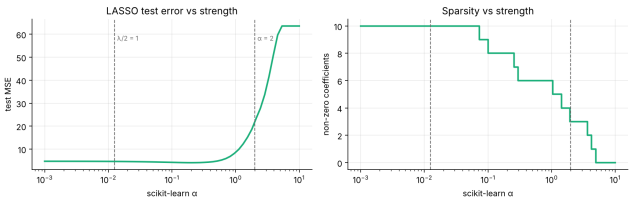


Figure 4. LASSO test MSE and number of non-zero coefficients as the penalty grows; dashed lines mark the correctly mapped penalty and $\alpha = 2$.

Next steps. (i) Repeat each experiment over many random splits and report the mean and spread of test error; (ii) choose λ by cross-validation rather than fixing it; (iii) compare gradient descent with Newton's method (IRLS), which converges in a handful of iterations for logistic regression; and (iv) report a precision–recall analysis for the imbalanced classification task.

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